KARACHI INSTITUTE OF ECONOMICS & TECHNOLOGY

SPRING-2022

Software Engineering

Artificial Intelligence (2+1)

Instructor: Muhammad Hussain
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Course Code: CS-4401
Pre-requisite(s): Introduction to Computer Programming (CS-1301) and Linear Algebra (MS-1302)
Faculty Contact Hours: Monday 11:30 am – 12:30 pm or by appointment

Objectives:

This course provides theoretical and practical knowledge of Artificial Intelligence to the students and enables them to implement and develop these algorithms. The objectives are to give a comprehensive understanding of artificial intelligence techniques and algorithms. It provides understanding of the basic issues of knowledge representation and blind and heuristic search, as well as an understanding of other topics such as game theory and minimax algorithm, etc. that play an important role in AI programs and also provides basic understanding of some of the more advanced topics of AI such as learning using Bayesian networks, decision trees and neural networks.

Learning Outcomes:

Mapping of CLOs and PLOs			
Sr. No.	**Course Learning Outcomes (CLOs)**	**Program Learning Outcomes (PLOs)**	**Bloom's Taxonomy**
CLO_1	Explain artificial intelligence, terminologies, its characteristics and its application areas.	PLO_1	C2
CLO_2	Solving search problems using various uninformed and informed search algorithms. Apply local search algorithms like Genetic Algorithm (GA) on optimization problems and perform Minimax search on games.	PLO_2	C3
CLO_3	Implementing various machine learning concepts including data handling, attribute subset selection, feature extraction, regression and classification.	PLO_3	C3

Assessment Heads (AH)	Number of Assessments	CLO 1 Portion	CLO 2 Portion	CLO 3 Portion	Total Marks	Assessment %
Quiz	3	(4%)	(4%)	(5%)	13	13%
Assignment	3	(4%)	(6%)	(5%)	12	12%
Mid Term	1	(15%)	(10%)	(0%)	25	25%
Final	1	(10%)	(10%)	(30%)	50	50%
Total	**12**	**(22%)**	**(35%)**	**(43%)**	**100**	**100%**

Text Book:

- Artificial Intelligence: A Modern Approach, by S. Russell and P. Norvig, 3rd Edition, Prentice Hall, 2009.

Reference Books:

- Artificial Intelligence: A Systems Approach, by M. Tim Jones, Infinity Science Press, 2008.
- Pattern Classification, by Richard O. Duda, Peter E. Hart and David G. Stork, Wiley Inter-science, 2nd edition, 2006.
- Pattern Recognition and Machine Learning, by Christopher Bishop, Springer, 2006.

<u>Course Outline:</u>

Week No.	Session No.	Contents	CLOs
1	1	Introduction to Artificial Intelligence	CLO-01
	2	Turning test	CLO-01
2	3	Agents, PEAS Model	CLO-01
	4	Rationality, Nature and properties of environment	CLO-01
3	5	Structures of agents	CLO-01
	6	Problem solving by searching, Uninformed search strategies	CLO-02
4	7	Breadth first search (BFS), Uniform Cost search	CLO-02
	8	Depth first search (DFS)	CLO-02
5	9	Depth limited search	CLO-02
	10	Iterative deepening DFS	CLO-02
6	11	Informed search strategies	CLO-02
	12	Greedy best first search	CLO-02
7	13	A* search	CLO-02
	14	Local search algorithms	CLO-02
8	15	Genetic algorithms (GA)	CLO-02
	16	Adversarial search	CLO-02
9	17	Games, Minimax algorithm,	CLO-02
	18	Alpha-beta pruning	CLO-02
10	19	Introduction to Machine Learning,	CLO-03
	20	Feature extraction and classification	CLO-03
11	21	K-nearest neighbor classifier	CLO-03
	22	Training and testing error, Confusion matrix, Sensitivity and Specificity	CLO-03
12	23	Bayesian Networks	CLO-03
	24	Naïve Based Bayes Models	CLO-03
13	25	Gaussian distribution, Covariance matrix	CLO-03
	26	Neural networks	CLO-03
14	27	Single layer Perceptron	CLO-03
	28	Introduction to non-metric method and decision trees	CLO-03
15	29	Entropy impurity, CART algorithm	CLO-03
	30	Query selection	CLO-03